

RESEARCH ARTICLE

Thermodynamics and Molecular-Scale Phenomena

Experimental solid–liquid equilibria and solid formation kinetics for carbon dioxide in methane for LNG processing

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Abstract

In cryogenic liquefaction processes, CO₂ poses a solid-formation risk even in trace concentrations. We present solid–fluid equilibrium (SFE) data for CO₂ in liquid methane at CO₂ concentrations from (52 to 500) ppm, extending the available data and indicating that models tuned to existing data over predict the solubility of CO₂ at LNG storage temperatures (~112 K) by nearly a factor of 3. The new data are used to improve the SFE model in the ThermoFAST software package. The formation kinetics of CO₂ solids in liquid methane are elucidated at conditions relevant to cryogenic gas processing. Repeated, ramped-temperature formation measurements provide a statistical basis for quantifying solidification risk. Nucleation rates extracted from the ramped-temperature data, consistent with those measured at fixed temperature, were used to extract parameters describing CO₂ solid formation in methane. These results significantly improve the ability to predict CO₂ solid formation risk in cryogenic natural gas processing.

KEYWORDS

cryogenic engineering, hydrocarbon processing, nucleation, phase equilibrium, thermodynamics/classical

1 | INTRODUCTION AND BACKGROUND

The formation of solids from impurities in natural gas presents a risk to the safe and reliable operation of cryogenic processes. Carbon dioxide is an impurity commonly present in natural gas and, depending on its concentration, can be thermodynamically stable as a solid at temperatures greater than typical process operating conditions.^{1,2} For this reason, the solubility of carbon dioxide in liquefied natural gas (LNG) mixtures has been studied for nearly 70 years. In spite of this, solid–fluid equilibrium (SFE) predictions of trace carbon dioxide in methane by thermodynamic models used in process simulations are

often inconsistent with available experimental data,³ especially at very low carbon dioxide concentrations, where few data are available. In fact, no solubility data have been reported for carbon dioxide concentrations below 172 ppm,⁴ which is more than 3 times larger than the standard industrial tolerance specification for LNG production (~50 ppm).⁵ Consequently SFE predictions in this process-critical region require extrapolation of empirically tuned thermodynamic models.³

While quality thermodynamic data are crucial to understanding the solid-stability conditions for impurities present in natural gas liquefaction, the quantification of the kinetics of solid formation is also

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important for estimating blockage risk.^{6–9} Solid nucleation is a stochastic process; the formation temperature and/or the time to form (induction time) can only be described using probabilistic models.¹⁰ Formation probability is dependent on the driving force for solid formation, formally quantified by the supersaturation, $\Delta\mu$; however, it is conventional to use subcooling, ΔT , as an experimentally measurable driving force given that supersaturation is proportional to subcooling at constant pressure.^{10–12} The subcooling at formation is defined as the difference between the thermodynamic equilibrium temperature, T_{eq} , and the temperature at which formation occurs, T_{form} :

$$\Delta T = T_{eq} - T_{form} \quad (1)$$

where T_{eq} corresponds to the melting (or liquidus) temperature at the same pressure as the formation condition.

Operationally, it can be useful to differentiate between melting and freezing (formation) temperatures. For example, it is possible to operate a process just below the SFE temperature (that is, at very small subcooling) without observing the formation of solids despite those solids being thermodynamically stable. This is because the low driving force at the specified process condition allows for nucleation induction times (time to formation) that are very long compared with the fluid's residence time in the process equipment. Reliable probabilistic models based on experimental solid formation data allow solid formation risk to be predicted for given operating conditions. Unfortunately, to our knowledge there are no openly available data describing the kinetics of solid carbon dioxide formation in liquefied natural gas systems. Collecting such data accurately requires a method capable of reliably detecting minute quantities of solid forming from low-concentration mixtures across the p - T space relevant to LNG production.

The available literature data for the formation of non-aqueous solids in cryogenic gas processing, including those used in this work for CO_2 in CH_4 , are available online as part of the Joint Expert Speciation System (JESS) Hydrocarbon Solubility Database.^{13,14} Since the work of Donnelly and Katz in the early 1950s, many attempts to measure SFE for the carbon dioxide + methane binary system have been made. A variety of techniques have been applied by different authors over this time, covering a wide range of experimental conditions. Approximately 53% of all data points from the literature were measured using analytical methods^{1,2,4,15–22} and 32% using synthetic methods.^{8,23,24} For the remaining 15% of reported data, the methods used were not detailed in the respective paper.^{25–29} For further information regarding phase equilibrium measurement methods, see the series of reviews published by Dorhn and co-workers.^{30–33} Significantly, despite the large number of data available for this binary system, none are reported unambiguously at conditions where liquid methane is present. Many data reporting a CO_2 concentration and a measured SFE temperature do not report an experimental pressure. Those data where SFE temperatures are provided together with a corresponding pressure measurement are at conditions either: (i) below the methane vapor–liquid equilibrium (VLE) curve (as plotted in the pressure–temperature plane); (ii) along the methane VLE curve; or

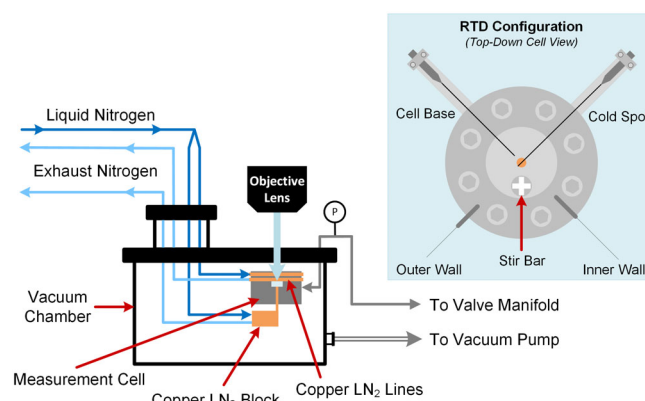


FIGURE 1 Schematic of the experimental apparatus used in this work, showing the measurement cell and vacuum chamber (left), with top-down view of the cell shown as an inset (right). Inset indicates the relative positions of the four thermometers (RTD) located in the cell body. Figures not to scale.

(iii) beyond the critical point of methane (supercritical). As a process primarily concerned with the production of liquid methane, the lack of unambiguous solid–liquid equilibrium (SLE) data, particularly at high pressures, is a major deficiency from the perspective of LNG production.

In this study, we use a synthetic, visual technique with high optical resolution, to detect the formation and melting of micron-scale carbon dioxide solids in liquid methane at carbon dioxide concentrations between 52 and 500 ppm. New SFE data are presented that enable the model predictions within the ThermoFAST software package to be made more accurate via regression.^{9,14,34,35} Measurements of CO_2 solid formation kinetics under ramped temperature conditions are presented and analyzed using the hazard-function method to extract subcooling-dependent nucleation rates.³⁶ These rates compare well to those extracted from slower fixed-temperature induction time measurements. The Classical Nucleation Theory (CNT) framework is used to explore the mechanisms governing the observed nucleation rates and, in particular, their dependence on concentration. Ultimately, the quantification of nucleation rate as a function of operational process variables allows the development of probabilistic models for solid formation risk, which in turn facilitates LNG production that is more reliable and efficient.

2 | METHODS

2.1 | Apparatus

The experimental apparatus used to gather the data presented in this work is described in detail by Sampson et al.⁷ and shown schematically in Figure 1. It consists of a cryogenic pressure cell with a sapphire window located at the center of the cell lid, permitting visual access to the cell via a long-working distance stereomicroscope (Leica, M205A). A 3 mm wide copper cold spot is incorporated into the base

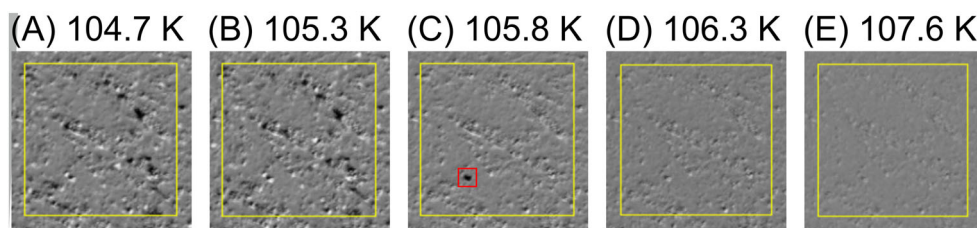


FIGURE 2 Example images from a stepped-melting experiment with 52 ppm CO₂ in CH₄ at 10.1 MPa. Each panel shows a 110 μm × 110 μm square region of interest (yellow square). Each panel shows a registered, contrast-enhanced image of a region of the cold spot, from which a reference image has been subtracted. The average temperature for the step on which each image is taken is indicated above each panel. End-of-melting occurs between panels (C) and (D); the last visible solid is indicated by the red square in panel (C).

of the cell; the copper is flush with the internal surface of the cell base and is cooled relative to the rest of the cell to promote nucleation on the cold spot. During the collection of data presented in this work, the low-temperature pressure seal was improved by replacing the trapezoidal glass filled PTFE o-ring described in Reference 7 with an indium seal. This new seal consists of 1.6 mm diameter indium wire, bent into a ring and laid in a groove cut in the cell body surrounding the sample well. This groove has a triangular cross section, and the indium seals against the face of the sapphire window when the lid is attached to the body of the cell.

Microscope images for this work were collected at a magnification of 50× with a digital microscopy camera (Leica, DMC5400), yielding a circular field-of-view (FOV) of ~3.5 mm diameter with the cold spot at its center. Image resolution varied between (5 and 20) megapixels (MP) depending on the experiment. All melting experiments used 20 MP images, as did all the formation measurements at concentrations of 500, 66, and 52 ppm CO₂. Formation measurements at 100 ppm used both 5 and 20 MP images, and there was no discernable difference in the subcoolings measured for this mixture using the different image sizes. The maximum resolution of the stereomicroscope is 525 line-pairs per millimeter, yielding a minimum resolvable particle size of 0.95 microns. The smallest solids detected in this work are 5–10 μm laterally, roughly equivalent to a volume of 35–260 μm³ if each solid is assumed to be a hemisphere with a radius half the lateral size. For carbon dioxide solids with no porosity, this would correspond to $(1.2 \text{ to } 9.5) \times 10^{-6}$ μmol per particle, assuming a solid density of 1.6 g·cm⁻³ estimated from the measured pressures and temperatures using the equation of state for solid CO₂ of Trusler.³⁷

Microscope images are matched to the pressure and temperatures recorded for the cell using purpose-written Python scripts that analyze image time stamps together with pressure and temperature time series data. Images are processed using FIJI software to perform rigid image registration (translation only), reference image subtraction, and contrast enhancement.³⁸ Figure 2 shows a set of processed images for a 110 μm² region-of-interest (ROI) containing carbon dioxide solids during a stepped melting experiment for a 52-ppm carbon dioxide in methane mixture. The image subtraction process results in solids being indicated by areas of maximum or minimum intensity (entirely white or entirely black regions, respectively). In contrast, areas containing both light and dark shading (i.e., area with one side

TABLE 1 Sample information for fluid mixtures used in this work

Substance	Composition	Source
Ultra-high purity CH ₄	99.9995 mol% ^a	CoreGas Australia
500 ppm binary mixture	(500 ± 5.0) ppm CO ₂ in CH ₄	CAC Gas & Instrumentation
100 ppm binary mixture	(100 ± 1.0) ppm CO ₂ in CH ₄	CAC Gas & Instrumentation
66 ppm binary mixture	(65.8 ± 0.9) ppm CO ₂ in CH ₄	Prepared in-house
52 ppm binary mixture	(52.2 ± 0.6) ppm CO ₂ in CH ₄	Prepared in-house

Note: Binary mixtures prepared in-house are volumetric dilutions of the 100 ppm CAC Gas mixture with the UHP methane from CoreGas.

^aManufacturer's stated impurities: <2 ppm H₂O; <2 ppm N₂; <1 ppm other hydrocarbons; <0.5 ppm O₂; <0.1 ppm H₂; no specification reported for CO₂ content.

that is consistently light, while the opposite side is consistently dark) are surface defects not completely removed due to imperfect image registration (caused by thermal expansion) and the effect of stirring the fluid. The melting temperature reported for each experiment is taken to be the average of the temperatures on the step when the last solid disappeared (i.e., where no solid is visible) and that of the preceding step. For the experimental run shown in Figure 2, the final solid particle melts between panels (C) and (D), yielding $T_{\text{melt}} = (106.0 \pm 0.25)$ K, where the uncertainty bound indicates ½ the step size used in the melting experiment. The uncertainty of the average melting temperature (T_{meas}) reported for each mixture includes the step size, repeatability of the measurements, and uncertainty in the thermometry chain, and is described in more detail with the results of the equilibrium measurements.

2.2 | Materials

Samples used in this work are listed in Table 1. Measurements of (100 and 500) ppm carbon dioxide in methane mixtures were conducted with samples prepared by CAC Gas & Instrumentation with a 1% relative uncertainty in the stated carbon dioxide mole fraction. To

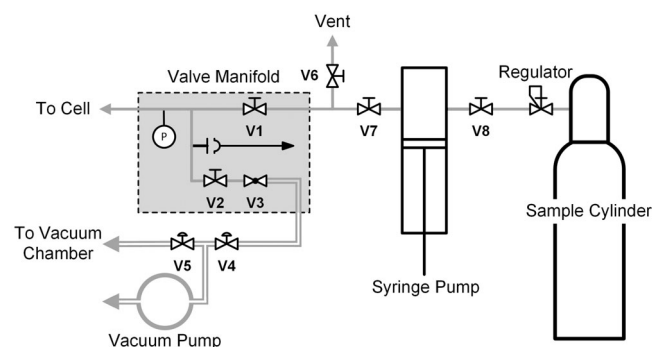


FIGURE 3 Apparatus diagram showing valve manifold and vacuum configuration (left) and sample dosing system (right). Figure not to scale.

prepare the (52 and 66) ppm mixtures, the purchased 100-ppm sample was diluted with 99.9995 mol% methane purchased from CoreGas Australia. Dilutions were prepared volumetrically in the syringe pump at pressures and temperatures above the critical point of methane. To prepare dilutions, the 100 ppm sample cylinder was connected to the syringe pump via a two-stage high pressure regulator and 1/8" o.d. stainless steel tubing. Figure 3 depicts this arrangement.

The syringe pump and tubing were then evacuated while the cylinder was heated to approximately 45 °C. Referring to Figure 3, valves V5, V6, and the cylinder valve (not shown) were closed during this step, while all other valves remained open. Once the evacuation step was complete, valves V2–V4 and V7 were closed, and V5 and V8 were opened. Sample from the mixture cylinder was then introduced at a specified pressure to the syringe pump using the two-stage regulator. The internal volume of the pump and the target pressure were based upon thermodynamic predictions made using the recommended equations of state (EoS) implemented in REFPROP 10.0.^{39–41} The pump was then isolated from the 100 ppm sample cylinder by closing the regulator and valve V8, and the 99.9995 mol% methane cylinder was attached to the regulator. The volume of the syringe pump was increased to the full capacity (266 ml) and valve V8 and the regulator were used to introduce methane into the pump to a predetermined pressure calculated to give the target carbon dioxide concentration while ensuring sufficient pressure and sample quantity to conduct the planned measurements. After mixture preparation was complete, valve V8, the regulator, and the valve on the cylinder itself were closed, and V7 was slowly opened to pressurize the cell. The final composition of the two dilutions prepared using this method and the purchased materials are reported in Table 1 together with the associated uncertainties in their composition. For the purchased binary mixtures, the compositional uncertainty given by the supplier is used. For the sample dilutions prepared in the lab, relative uncertainty in composition was estimated for each dilution from the measured variables used to calculate mixture composition (p, V, T) and the stated compositional uncertainty of the starting materials. Compositional uncertainty in terms of parts-per-million for each dilution was then determined by multiplying the relative uncertainty in mixture composition by the calculated number of moles of carbon dioxide.

2.3 | Measurement technique

The techniques used for determining onset of solid formation and SFE conditions are the same as those described in Reference 7 for benzene + methane mixtures. Solid carbon dioxide was formed on the cell's cold spot during isobaric constant-cooling ramps as well as during fixed temperature experiments. During formation experiments, the temperature of the cold spot was (1–2) K colder than the surrounding cell base and walls to promote growth in the microscope's field-of-view. To ensure good mixing of the fluid without compromising image quality, the measurement cell was stirred for the duration of each experiment at 300 RPM for 20 s with a 10 s pause between each stirring period. Cooling during ramped formation experiments, and subsequent heating to prepare for the following experiment, was conducted at a target rate of $\beta = 1 \text{ K} \cdot \text{min}^{-1}$. Actual average ramp rates for each mixture composition are provided in the Supplemental Information, as some experiments did not meet the target cooling rate. These average ramp rates were used to conduct the nucleation rate analysis detailed in the discussion of the kinetic data. Failure to meet to target ramp rate was due to either insufficient liquid nitrogen flow, or because formation did not occur until the low-temperature limit of the apparatus was approached. The average ramp rates reported represent the cooling rate achieved after entering the SFE region and onset of solid formation across all ramped experiments in the dataset. Apart from the 52 ppm dataset, for which several formation events occurred at constant temperature, most of the data were unaffected by any deviation from the target ramp rate. Further discussion of the 52 ppm data can be found in the discussion section and in the Supplemental Information. Excluding the experiments where formation occurred at constant temperature, the average ramp rate across all concentrations was $\beta_{\text{avg}} = (0.9 \pm 0.2) \text{ K} \cdot \text{min}^{-1}$.

For solid formation experiments conducted at constant temperature, the cell was first cooled at a rate of approximately $2 \text{ K} \cdot \text{min}^{-1}$ to the target temperature where it was held to await formation. These experiments were conducted only with the highest concentration mixture, and consequently were far from the low-temperature limit of the apparatus. The average time spent on the ramp while in the SFE region was (6 ± 2) minutes before the cell stabilized at the target temperature.

To determine SFE conditions, particles formed during constant cooling were melted by increasing the temperature of the cell in steps of (0.5–1.0) K, with each step held for at least 1 h. Referring to the inset of Figure 1, three of the RTDs in the cell measure surfaces that are in contact with the fluid. The cell base and cold spot RTDs are parallel to and just below the bottom surface of the cell, while the inner wall RTD measures the temperature of the cell wall near the fluid. During melting experiments the temperature differences between these surfaces were reduced to <1 K for each step, with the exception of one of the two 66 ppm melting experiments (2 K difference); the larger temperature difference is reflected in the higher uncertainty of the corresponding T_{meas} value reported in Table 2. Images were collected using the stereomicroscope at 50 $\times$ magnification every 30 s during ramped experiments and every 2 min during stepped melting. A total of two melting experiments were conducted in this way for

TABLE 2 Measured solid–liquid equilibrium data for the carbon dioxide + methane binary system

$x_{\text{CO}_2}/\text{ppm}$	$p_{\text{meas}}/\text{MPa}$	T_{meas}/K	Steps	Range/K
500	10.00	122.8 ± 1.8	1 K; ≥ 2 h	114.2–123.3 K
	6.98	121.4 ± 1.8	1 K; ≥ 2 h	117–124 K
100	10.09	111.3 ± 1.9	1 K; ≥ 1 h	100–117 K
	7.03	111.0 ± 1.9	1 K; ≥ 1 h	98.5–112.5 K
65.8	10.10	108.4 ± 2.4	1 K; ≥ 1 h	100–116 K
52.2	10.09	106.3 ± 1.7	0.5 or 1 K; ≥ 1 h	100–111 K

Note: All experiments were carried out with constant temperature steps and image collection every 120 s. The spacing, duration and temperature range of the steps are indicated in the table. The reported T_{meas} for the 66 ppm mixture is the average of two separate measurements. The value reported for the 52 ppm mixture is the average of three measurements. Standard uncertainty in pressure is 0.08 MPa.

the 66 ppm binary mixture; the reported condition is an average of these values. In the same way, three measurements were conducted for the 52 ppm binary mixture. A single measurement each at pressures of 7 and 10 MPa were recorded for the 100 and 500 ppm carbon dioxide + methane samples.

3 | RESULTS AND DISCUSSION

3.1 | Solid–liquid equilibrium data

A total of nine stepped-melting experiments were conducted to determine the SFE temperatures at six different p - x conditions for the carbon dioxide + methane binary system via the synthetic method. The resultant values are tabulated as T_{meas} in Table 2.

The primary sources of uncertainty for these measurements are the uncertainty in composition and the uncertainty in temperature. The compositional uncertainties are provided in Table 1 and are discussed in the materials section. The uncertainty in measured SFE temperatures is estimated based on step size, repeatability of measurements, uncertainty in the measured temperature (i.e., based on the thermometry chain, calibration results, and the difference between the three RTD readings at the melting conditions). The individual uncertainty contributions are combined in quadrature. The resulting combined standard ($k = 1$) uncertainty values are summarized in Table 2. For more information on the uncertainty analysis for this apparatus, see Reference 7.

The new SFE data are plotted in Figure 4, alongside available literature for the solubility of CO_2 in CH_4 ^{4,17–19,24,26} and the corresponding solubility predictions of ThermoFAST (discussed in the following section). Below 1000 ppm carbon dioxide in methane, the literature data lie almost exclusively to the left of the current measurements. This may reflect the limited detection sensitivity of previous measurements, since higher resolution techniques are able to reveal the presence of any smaller solids which persist at higher temperatures. Most data in the literature are derived from analytical SFE techniques in which the overall mixture composition is not necessarily known. Solids are formed in excess at a given p - T condition and the concentration of carbon dioxide in the fluid phase at equilibrium is determined by sampling and then analyzing the fluid composition, most commonly by gas chromatography. The sparseness of available solubility data for

trace concentrations of carbon dioxide in methane, particularly in liquid methane at high pressures, presents an opportunity for others to investigate this industrially relevant system further.

3.2 | Tuning of binary interaction parameters for SFE predictions

The new SFE data were used to improve the accuracy of the solubility predictions of the thermodynamic property models implemented in the software package ThermoFAST.^{34,35} A brief description of the models and tuning procedure involved is given below; further detail may be found in References 9 and 42. To predict the properties of fluid mixtures for SFE calculations, the default model used by ThermoFAST is the Peng–Robinson cubic equation of state⁴³:

$$p = \frac{RT}{v - b_m} - \frac{a_m}{v(v + b_m) + b(v - b_m)} \quad (2)$$

where p is pressure, $R = 8.3144626 \text{ J}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}$ is the gas constant, T is temperature, and v is molar volume. The parameters a_m and b_m are averages over the mixture components calculated using the van der Waals mixing rules:

$$a_m = \sum_i \sum_j x_i x_j (1 - k_{ij}) \sqrt{a_i a_j}$$

$$b_m = \sum_i x_i b_i \quad (3)$$

where x_i is the mole fraction of component i , and k_{ij} are the binary interaction parameters. The pure component parameters a_i and b_i are given by:

$$a_i = \left[1 + m_i \cdot \left(1 - T_{r,i}^{1/2} \right) \right]^2 \alpha(T_{r,i}, \omega_i) \cdot 0.45724 \frac{R^2 T_{c,i}^2}{p_{c,i}}$$

$$b_i = 0.07779 \frac{RT_{c,i}}{p_{c,i}}$$

$$m_i = 0.37464 + 1.5422\omega_i - 0.26992\omega_i^2 \quad (4)$$

where $T_{r,i} = T/T_{c,i}$ is the reduced temperature for component i , while $T_{c,i}$, $p_{c,i}$ and ω_i are the critical temperature, critical pressure, and

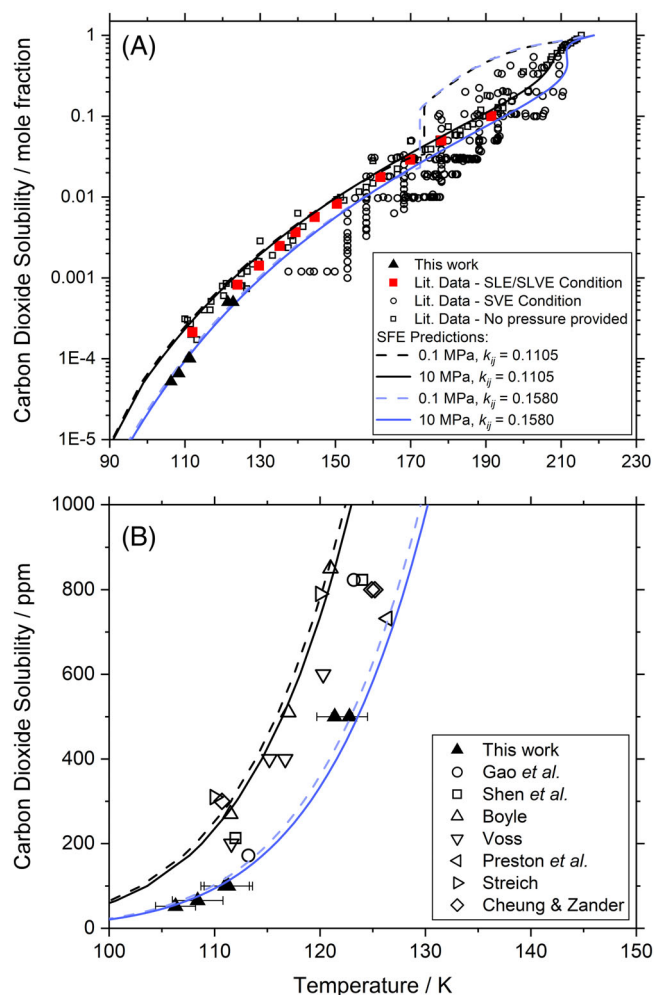


FIGURE 4 Literature and experimental data from this work for the solubility of carbon dioxide in methane. Error bars in panel (B) indicate the combined standard ($k = 1$) uncertainty in the temperature measurement (see Table 2). Uncertainty in composition is contained within the size of the plot symbols (see Table 1). Model predictions at 0.1 and 10 MPa are generated using ThermoFAST^{34,35} with the binary interaction parameter (k_{ij}) set to either the previous default value (0.1105) or the tuned value determined using the data measured this in work (0.1580).

acentric factor, respectively. Equations (2)–(4) allow the (partial) fugacity of each component in the fluid mixture to be calculated at a specified temperature, pressure, and composition.

To calculate the fugacity of a pure solid phase formed by component i , the so-called “classical” model is used⁴⁴:

$$\ln \phi_{\text{pure},i}^S = \ln \phi_{\text{pure},i}^L - \frac{\Delta H_{f,i}}{RT_{m,i}} \left[\frac{T_{m,i}}{T} - 1 \right] + \frac{\Delta c_{p,i}}{R} \left[\frac{T_{m,i}}{T} - 1 + \ln \left(\frac{T}{T_{m,i}} \right) \right] - \frac{\Delta v_i(p - p_{\text{atm}})}{RT} \quad (5)$$

Here p_{atm} is atmospheric pressure, $\phi_{\text{pure},i}^L$ is the fugacity of component i in the (pure) liquid phase, $T_{m,i}$ is the melting temperature, $\Delta H_{f,i}$ is the heat of fusion, $\Delta c_{p,i}$ ($= c_{p,i}^L - c_{p,i}^S$) is the change of the isobaric

heat capacity from solid to liquid phase and Δv_i is the change in molar volume from solid to liquid phase.

For certain compounds, like CO_2 and benzene, the pure solid fugacity can also be calculated using the Helmholtz equations of state of Trusler³⁷ and Xiao et al.,⁴⁵ respectively, both of which are implemented in ThermoFAST. However, use of the Trusler EoS for pure solid CO_2 instead of Equation (5) did not improve the root mean square (RMS) deviation between the predicted and calculated SFE temperatures.

The tuning procedure consists of adjusting the fluid phase binary interaction parameters k_{ij} to minimize the difference between calculated SFE conditions and the experimental data for binary mixtures containing components i and j . The objective function used to tune the ThermoFAST model is the RMS error:

$$\text{RMS}_T(k_{ij}) = \sqrt{\frac{\sum_{l=1}^N [T_{\text{exp},l} - T_{\text{calc},l}(k_{ij})]^2}{N}} \quad (6)$$

where N is the number of experimental data points, $T_{\text{exp},l}$ are the experimentally measured melting temperatures and $T_{\text{calc},l}$ are the corresponding temperatures calculated using k_{ij} . The value of k_{ij} which minimizes $\text{RMS}_T(k_{ij})$ is determined to within a tolerance of 0.005 K.

Carrying out the tuning using the measurements from Table 2, we obtained the value $k_{ij} = 0.1580$ with an associated RMS error of 1.07 K for $\text{CO}_2 + \text{CH}_4$ mixtures. The previous value of the binary interaction parameter was $k_{ij} = 0.1105$ and curves showing solubility calculations obtained with these two values are included in Figure 4. Over the full range of temperatures (Figure 4A) the two sets of curves both lie within the spread of measured values. The two sets of model predictions differ most noticeably at low temperatures, where the new binary interaction parameter has been forced to fit the high-resolution experimental data reported here. Both models (tuned and untuned) predict that solubility increases as the pressure decreases, although below ~ 170 K the pressure dependence is negligible. The insensitivity to pressure predicted by the model suggests that much of the data scatter, particularly for points indicated as reflecting solid-vapor equilibrium, may suffer large experimental uncertainties.

The untuned k_{ij} value (0.1105) represents all data reported with experimental pressures (including the data reported in this work) with an RMS error of 8.22 K. The tuned value (0.1580) represents the same data with an RMS error of 5.52 K. A prior exercise in tuning the ThermoFAST model by Baker et al. excluded many of the higher-temperature SVE conditions; the primary objective was to improve predictions for LNG production at industrially relevant conditions.^{9,42} The binary interaction parameter resulting from the work of Baker et al. ($k_{ij} = 0.3136$) represents the subset of data used in the tuning with an RMS error of 2.06 K. Since it is unlikely that a single binary interaction parameter can be found which works well across the entire range of temperatures, pressures, and compositions, this type of model tuning should be done in the context of a specific goal. The problem is compounded by the variability in data measured at similar conditions. For this reason, new data, together with robust uncertainty estimates, are needed at the low- CO_2 content, liquid methane

TABLE 3 Summary of subcooling, ΔT , at formation for the constant cooling ($1 \text{ K} \cdot \text{min}^{-1}$) experiments with each carbon dioxide + methane binary mixture

	52 ppm	66 ppm	100 ppm	500 ppm	Total
Total number of experiments	33	15	48	25	121
Number of formation events	21	15	48	25	109
Number of experiments with no formation	12	0	0	0	12
ΔT range / K	6.3–12.9	11.3–13.7	9.2–17.5	6.3–14.6	6.3–17.5
Average ΔT / K	10.3 ± 1.6	12.6 ± 0.7	13.3 ± 1.8	11.8 ± 1.6	12.3 ± 1.9

Note: The bounds reported for the average subcooling correspond to the standard deviation of the data. All 109 formation events are shown in Figure 5.

conditions measured in this work. The tuned value of $k_{ij} = 0.1580$ determined here improves the overall RMS error for all data (reduction from 8.22 to 5.52 K) as well as improving predictions at low-temperature, LNG-relevant conditions. The new optimized value of k_{ij} has been set as the default value for SFE calculations involving mixtures with $\text{CO}_2 + \text{CH}_4$ in both the desktop version of ThermoFAST and the online application, ThermoFAST Web.^{34,35}

3.3 | Kinetic data

Having established thermodynamic equilibrium conditions for the low-concentration carbon dioxide in liquid methane samples, ramped cooling experiments were conducted with the goal of determining the formation kinetics of this binary system at high-pressures and cryogenic temperatures. Independent formation onset events measured along constant cooling ramps under isobaric conditions were recorded for all 4 mixture compositions at pressures of $p = (4, 7, \text{ and } 10) \text{ MPa}$. Table 3 summarizes the formation data, which are shown in Figure 5.

The formation subcoolings for the experiments shown in Figure 5 were sorted into linearly spaced bins to generate the formation probability histograms displayed in Figure 6. The low-concentration (52 and 66 ppm) mixtures were combined to facilitate subsequent comparisons with a predicted distribution determined from models fit to the combined 100 and 500 ppm data. When treated as a single dataset, the data from the low concentration mixtures are referred to as 59 ppm (the arithmetic mean of these concentrations). Bin widths were selected such that the resultant histograms exhibit a monotonic increase and decrease on either side of the mean subcooling, based on previous work in which greater numbers of formation events were used to generate the histograms.^{36,46,47} Accordingly, the choice of bin width, δT_f , varies slightly between each dataset: In Figure 6, (a) $\delta T_f = 1.5 \text{ K}$ (59 ppm), (b) $\delta T_f = 1.25 \text{ K}$ (100 ppm), and (c) $\delta T_f = 1.75 \text{ K}$ (500 ppm). Additionally, the 500-ppm bins begin from a value of 0.5 K rather than 0 K; since bins before the first formation event are irrelevant the application of an offset to the histogram's initial bin simply helps address discretization issues associated with sparse data. A full list of formation conditions for each dataset are provided in the Supplemental Information, which allow the original data to be used with various binning criteria for further analysis. The histograms shown in Figure 6 include all data listed in Table 3,

including the nine 52 ppm formation events which occurred at constant temperature and the 12 no-formation experiments from the same dataset. That is, Figure 6A represents 48 total experiments: 36 with formation and 12 without. Consequently, the resulting histogram does not sum to one, which reflects the observed formation probability.

In addition to the formation data measured along the constant cooling ramps, 34 solid carbon dioxide formation events at a constant subcooling of $\Delta T = (8.0 \pm 1.0) \text{ K}$ were recorded for the 500 ppm sample. A ramp rate of $2 \text{ K} \cdot \text{min}^{-1}$ was used to cool the system down to this constant temperature. As mentioned previously, slight slowing of the ramp rate as the system approached the final hold temperature resulted in an average time of $(360 \pm 120) \text{ s}$ between entering the SFE region and reaching stability at the final hold temperature. Figure 7A shows the induction time histogram generated from this set of formation events at constant temperature, while Figure 7B shows the cumulative formation probability distribution for induction times obtained directly from the histogram.⁴⁸ Logarithmically spaced bins were used to generate the histogram because using a linear-spacing produces many empty bins at long induction times, which artificially biases fits of Equation (7) to the resulting cumulative probability distribution function.^{49,50}

The nucleation rate, J , is a key parameter in quantifying solid-formation risk. Under isothermal conditions (constant subcooling), the cumulative probability of solid formation is a function of J and the induction time at the given subcooling. According to the mononuclear nucleation mechanism of Classical Nucleation Theory (CNT), formation at constant subcooling yields an exponentially distributed set of induction times, t^{10} :

$$F(t) = 1 - \exp(-Jt) \quad (7)$$

In Equation (7), $F(t)$ is the cumulative probability of solid formation at a given time, t . Equation (7) has been used successfully to represent measured distributions of induction times acquired at constant subcooling for gas hydrate formation.^{6,47,49–53} By fitting the induction times measured for the 500 ppm CO_2 in CH_4 mixture to Equation (7), the nucleation rate at this condition ($p = 7 \text{ MPa}$, $\Delta T = [8.0 \pm 1.0] \text{ K}$) was estimated to be $J = (2.2 \pm 0.2) \times 10^{-3} \text{ s}^{-1}$.

While nucleation rates at constant temperature can be determined by fitting induction time formation probabilities to Equation (7),

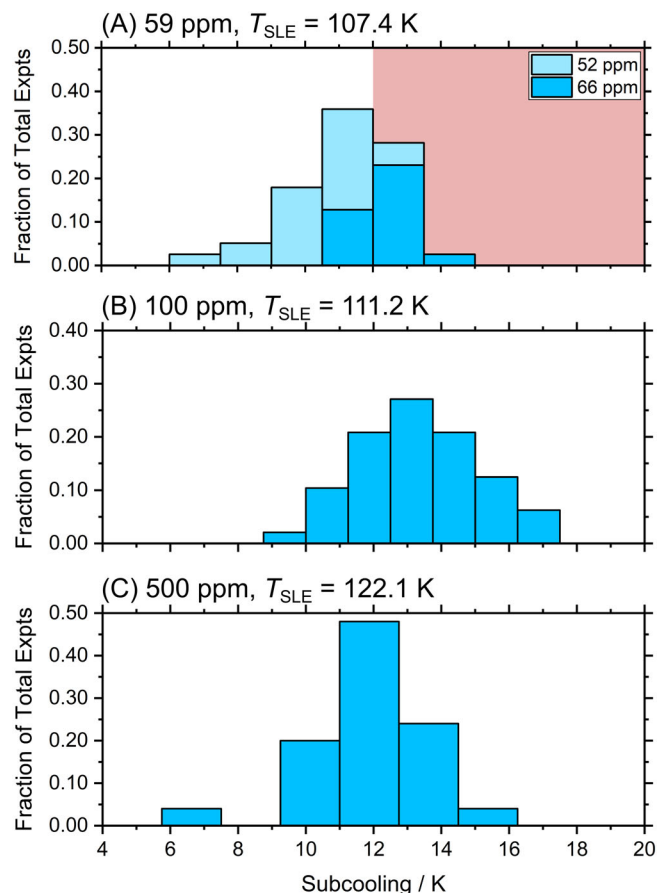
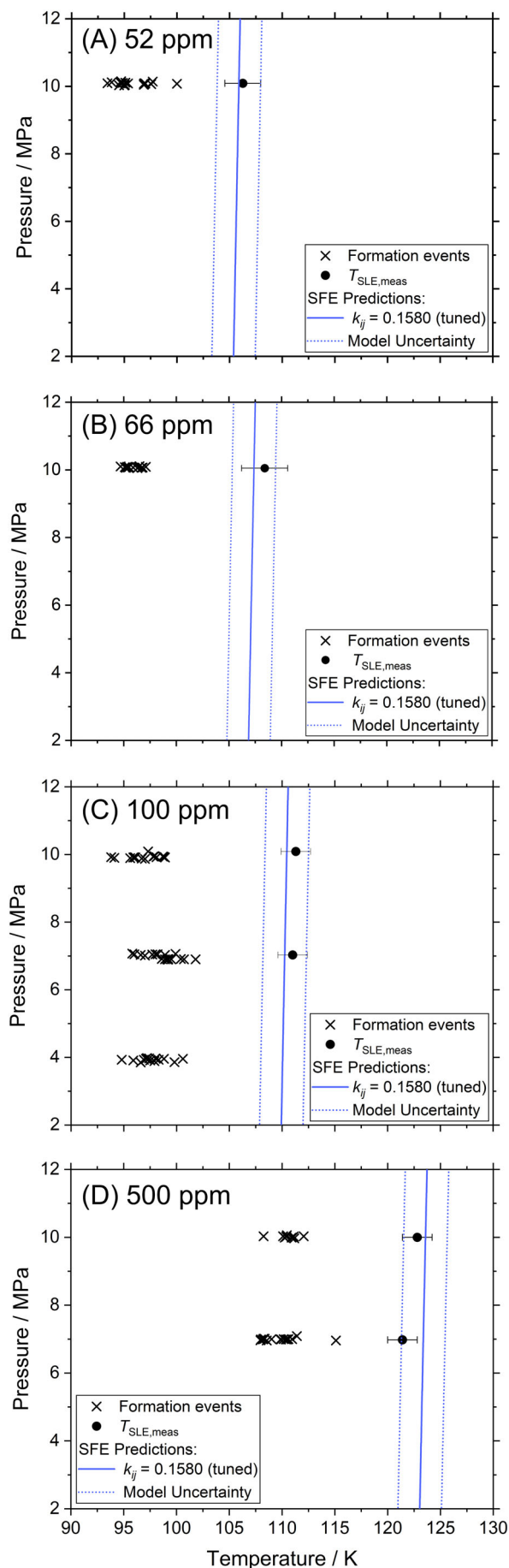


FIGURE 6 Probability of formation histograms for constant-cooling experiments in mixtures with (A) 52 or 66 ppm CO_2 in CH_4 , (B) 100 ppm CO_2 in CH_4 , and (C) 500 ppm CO_2 in CH_4 . Subcooling, ΔT , is calculated for each formation event according to Equation (1), where T_{eq} is the respective SFE temperature recorded in Table 2. The red shaded region in (A) indicates subcoolings corresponding to formation temperatures which are inaccessible at $1\text{ K}\cdot\text{min}^{-1}$ for the 52 ppm mixture using the current apparatus set-up.

such measurements are slow and limited in their ability to efficiently probe how J varies with subcooling. The measurement of solid formation along constant-cooling ramps can more efficiently provide information about the dependence of nucleation rate on subcooling if analyzed correctly.⁴⁶ Recently, Barwood et al.³⁶ demonstrated an efficient method of extracting methane hydrate nucleation rates from

FIGURE 5 Experimental melting temperatures (T_{SLE}) and formation temperatures for carbon dioxide solids in mixtures of (A) 52 ppm, (B) 66 ppm, (C) 100 ppm, and (D) 500 ppm CO_2 in CH_4 , measured under ramped temperature conditions at constant pressure. Error bars on T_{SLE} are indicative of the uncertainty bounds listed in Table 2. Predicted SFE boundaries were generated using ThermoFAST for each mixture with the binary interaction parameter tuned to the T_{SLE} data from this work. Model uncertainty is based on the root mean squared deviation stated by Baker et al. for the $CO_2 + CH_4$ binary system at LNG-relevant conditions (not including the data of this work) (2.06 K).^{9,42}

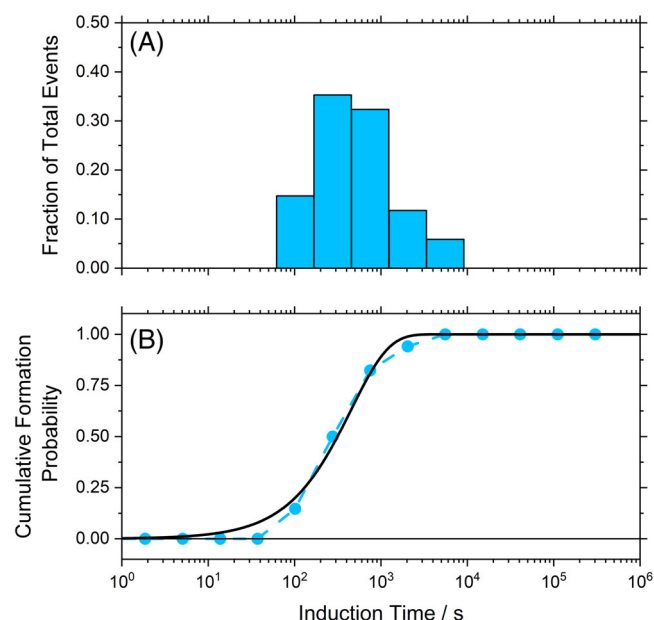


FIGURE 7 (A) Formation data at constant subcooling $\Delta T = 8$ K relative to the measured $T_{\text{SLE}} = 121.4$ K at 7 MPa for 500 ppm carbon dioxide in methane. (B) Cumulative formation probabilities (blue circles) calculated from the 34 formation events plotted as a function of induction time. The solid black curve indicates the fit of the data to Equation (7), which yields a nucleation rate of $J = 0.0022 \text{ s}^{-1}$.

formation data measured along constant cooling ramps that were equivalent to those determined by constant temperature experiments. The basis of this method is the hazard function, $h(t)$,^{54–56} typically used in survival analysis,⁵⁷ which accounts for the system's history in the solid stability region. For ramped-temperature formation experiments at a constant cooling rate, the hazard function can be written in terms of the subcooling at formation:

$$h(\Delta T) = \frac{f(\Delta T)}{1 - F(\Delta T)} \quad (8)$$

The numerator of Equation (8) is the formation probability density function (PDF) discretized into subcooling bins of equal width. The denominator is one minus the integral of the PDF, that is, the cumulative distribution function (CDF). All CDFs in this work were generated by summing the bin populations directly to determine the cumulative probability of having observed formation once the system has reached the subcooling at the right edge of each bin. This method avoids any (minor) artificial broadening of the distribution that can be generated during trapezoidal integration. Multiplying the hazard function by the cooling rate, β , yields a subcooling-dependent nucleation rate for every bin in which a formation event occurred.

$$J(\Delta T) = h(\Delta T)\beta \quad (9)$$

This method was used to obtain values of J for the $\text{CO}_2 + \text{CH}_4$ binary mixture from the formation data measured along constant

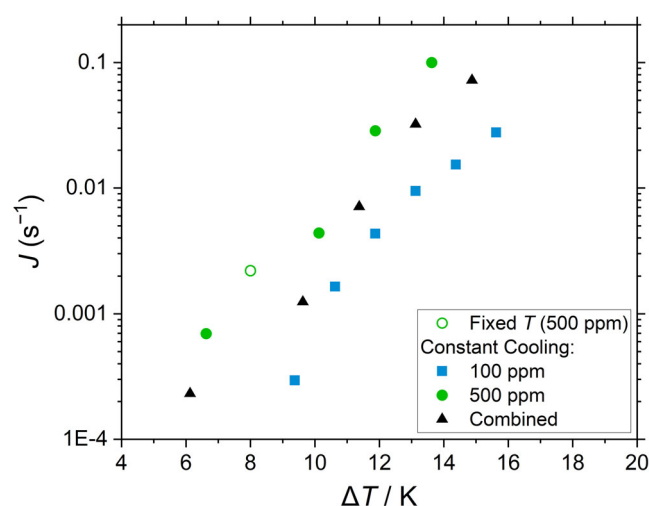


FIGURE 8 Nucleation rates, J , at different subcoolings from formation experiments conducted in this work for 100 and 500 ppm CO_2 in CH_4 mixtures. J values for constant cooling (ramped) experiments were determined by the hazard function method detailed in Reference 36. The nucleation rate measured at constant temperature for the 500 ppm mixture was determined by fitting Equation (7) to the fixed-temperature data set (34 formation events at $\Delta T = 8$ K).

cooling ramps for the combined 100 and 500 ppm data sets. It is useful to combine data sets to increase the number of formation events being analyzed and reduce the influence of statistical fluctuations if the probability distribution depends only on subcooling. However, the formation data acquired at 52 and 66 ppm could not be combined with the higher concentration data sets because the lower temperature limit of the apparatus (91 K) and the SLE curve for pure methane (90.9 K), limit the lowest subcooling that could be achieved in those experiments, particularly while maintaining a constant cooling ramp rate as required by the analysis method. Consequently, out of a total of 33 experiments measured at the lowest concentration (52 ppm), no formation event was recorded for 12 ramps ($\Delta T_{\text{max}} = 9.3 \pm 0.3$ K), and formation occurred during 9 experiments after the cell reached constant temperature ($\Delta T_{\text{avg}} = 9.7 \pm 1.2$ K). The 12 experiments with no formation still provide information about the probability distribution governing the formation kinetics and have been included in the total number of experiments used to construct the histograms shown in Figure 6. However, without measured ΔT values, the results of these no-formation experiments do not appear in Figure 5.

Accordingly, only the 100 and 500 ppm data were used to determine subcooling dependent nucleation rates using the hazard function method. A combined PDF for the 100 and 500 ppm mixtures was generated by discretizing the 73 subcooling at formation values into bins of $\delta T_f = 1.75$ K. From this distribution and its associated CDF, nucleation rates were determined according to the hazard function method. These nucleation rates for the combined dataset are plotted as a function of experimental subcooling alongside those determined separately for each concentration in Figure 8. The nucleation rate determined from the fixed-temperature data measured for the

500 ppm CO₂ in CH₄ mixture is also included in Figure 8. Good consistency is exhibited between the nucleation rates measured for the 500 ppm system using the two different experimental methods and analysis techniques.

The results in Figure 8 suggest that the 500 ppm CO₂ in CH₄ mixture may have a higher nucleation rate than the 100 ppm mixture. Classical Nucleation Theory suggests that a linear dependence on solute concentration may be observable¹¹; however, normalizing the data in Figure 8 by solute concentration reveals that the scatter within the two data sets is comparable with their separation. Therefore, while a concentration dependence may exist, it cannot be resolved from the present data measured at different concentrations.

CNT describes the nucleation rate in terms of the measured subcooling (ΔT) with the expression:

$$J = A \exp\left(\frac{\Delta s_e \Delta T}{k_B T}\right) \exp\left(-\frac{B'}{T \Delta T^2}\right) \quad (10)$$

In Equation (10), T is the formation temperature, k_B is the Boltzmann constant, and Δs_e is entropy of fusion for the solid-forming compound, which was calculated to be $9k_B$ for CO₂ solids using the Peng-Robinson EoS implemented in MultiFlash 6.2.⁵⁸ A and B' are the kinetic and thermodynamic nucleation parameters, respectively. According to CNT, A is related to the number of active heterogeneous nucleation sites (N_0) through the product of z , the Zeldovich factor ($0.01 < z < 1$), and f , the attachment frequency of monomers to the growing nucleus:

$$A = z f N_0 \quad (11)$$

The thermodynamic nucleation parameter, B' , is a measure of the nucleation work required at these active heterogeneous sites:

$$B' = \frac{4c^3 v^2 \sigma_{ef}^3}{27 k_B \Delta s_e^2} \quad (12)$$

In Equation (12), c is a shape factor, v is the volume of the monomer forming the solid, and σ_{ef} is the energy penalty for introducing the new solid–fluid interface. Equation (10) can be linearized and rearranged,^{11,36,47,59} taking the form:

$$\ln J - \frac{\Delta s_e \Delta T}{k_B T} = \ln(A) - \frac{B'}{T \Delta T^2} \rightarrow y = \ln(A) - B'x \quad (13)$$

such that $y = \ln J - \frac{\Delta s_e \Delta T}{k_B T}$ and $x = \frac{1}{T \Delta T^2}$. When plotted in this form, as shown in Figure 9, a linear trend in the measured nucleation rates indicates that the values of A and B' are constant over the corresponding range of subcooling. The J values used to construct Figure 9 are from the combined 100 and 500 ppm distribution.

A fit to the linearized nucleation rate data from the combined distribution is shown in Figure 9. The nucleation parameters determined from the linear regression to the data are $A = (2.2 \pm 1.9) \times 10^{-2} \text{ s}^{-1}$ and $B' = (2.2 \pm 0.6) \times 10^4 \text{ K}^3$. The nucleation parameters determined

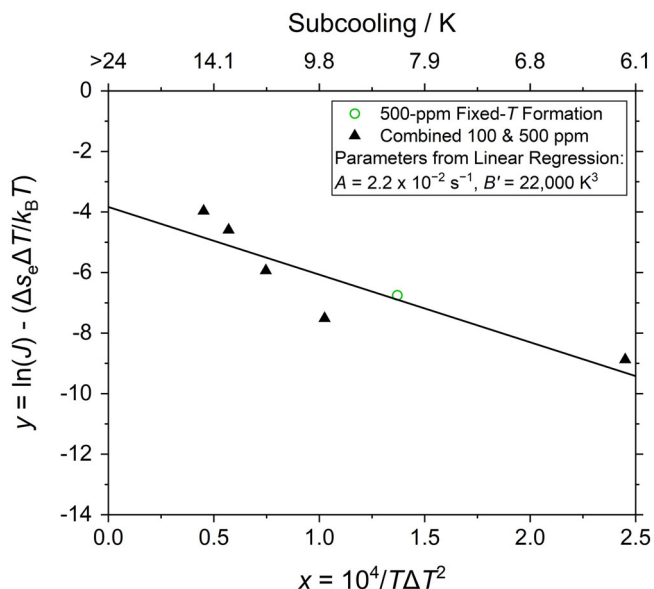


FIGURE 9 Nucleation rates linearized in the form of Equation (13) obtained from constant cooling and constant temperature formation data. Solid line is a linear fit to the nucleation rates extracted from the combined 100/500 ppm data.

from this fit can also be used to generate a formation distribution for comparison against the measured low concentration data. In Figure 10, the solid curve indicates the predicted CDF for the 59-ppm mixture based on the fit from the combined 100 and 500 ppm data. The points represent the measured CDF from the 59-ppm dataset. The experimental distribution does not reach its maximum value of one because no formation was observed for 12 measurements. The orange shaded area indicates temperatures that are at the lower limit of the apparatus' range, where cooling ramps of $1 \text{ K} \cdot \text{min}^{-1}$ are difficult to maintain, and the experimental CDF is likely impacted. The red shaded area indicates temperatures at or below the predicted methane T_{SLE} . While the predicted CDF continues into this region, it would not be possible after this point to distinguish carbon dioxide solids forming from the fluid phase, as the solvent itself begins to freeze. The CDF therefore effectively ends at the start of this shaded region ($T = 90.9 \text{ K}$), which is indicated by the dotted portion of the curve. This results in a prediction of a 98% probability that CO₂ solids will form during such experiments before the methane solid stability condition is reached. The CDF agrees well with the predicted curve, indicating that although the low-concentration data were not suitable for a robust statistical analysis, they do still provide information about formation kinetics in methane binary mixtures with low-CO₂ content.

The nucleation rates determined here for CO₂ in CH₄ can be compared with the likely residence time of impurities in the main cryogenic heat exchanger (MCHE) of an LNG plant to help contextualize freeze-out risk. The analysis of Baker et al.⁶⁰ of the 2014 RasGas incident reported by Ismail and Al Thani⁵ indicates that the residence time of fluid in the MCHE is about 30 s or less. The plant data reported by Ismail and Al Thani⁵ also indicated that the concentration of carbon dioxide entering the MCHE reached or exceeded 100 ppm

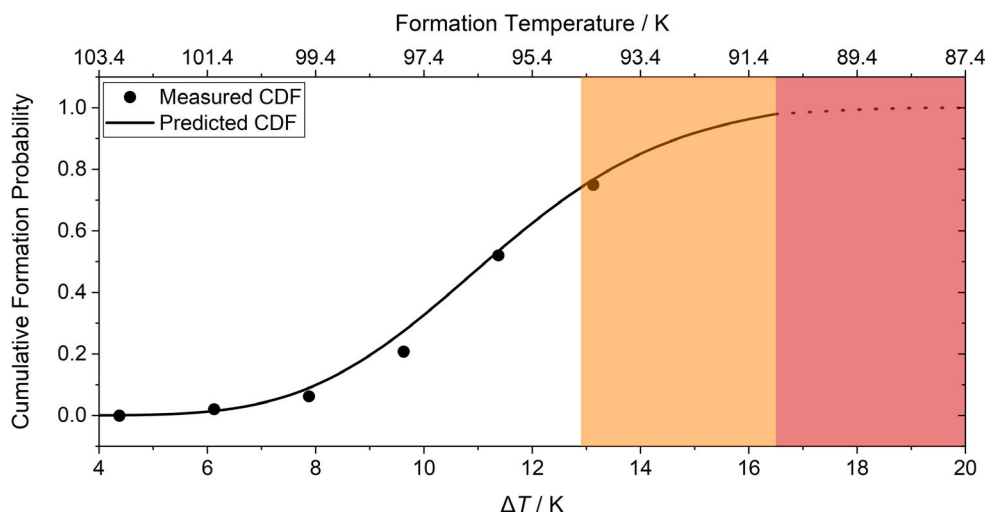


FIGURE 10 Cumulative probability distribution of formation events measured for the 59 ppm mixture (Figure 6A) compared with the cumulative distribution function predicted using Equations (9) and (10) together with the nucleation parameters extracted from the linearized data shown in Figure 9 for the 100 and 500 ppm mixtures. The orange shading indicates temperatures where the assumption of constant cooling rate required by Equation (9) is tenuous, while the red region corresponds to temperatures where the methane solvent could solidify.

(twice the industry-standard specification) on 14 separate occasions in <12 months, yet no CO₂ solid formation was evident. A 50% probability of CO₂ solid formation on the 30 s time scale of the MCHE residence time would require a nucleation rate of 0.033 s⁻¹. Referring to Figure 8, the range of subcooling for which this nucleation rate is possible is approximately (12 < ΔT < 16.5) K. Taking the lowest subcooling from this range gives the most conservative estimate for the CO₂ concentration corresponding to a nucleation rate of $J = 0.033 \text{ s}^{-1}$. Assuming that solid formation kinetics are largely independent of composition at the same subcooling, as observed in this work, to achieve this nucleation rate at a formation temperature of 125 K gives a required solid–fluid equilibrium temperature $T_{\text{SFE}} = 137 \text{ K}$. This equilibrium temperature can be combined with the plant operating pressure ($p \approx 5 \text{ MPa}^{60}$) to determine the concentration required to produce a 50% probability of CO₂ solid formation. The tuned model in ThermoFAST gives a value of $x_{\text{CO}_2} = 1960 \text{ ppm}$. This is nearly 50 times higher in concentration than the CO₂ content observed for the gas mixture entering the MCHE in the RasGas LNG plant, for which the predicted melting temperature is only 104 K. This analysis is consistent with the fact that no CO₂ freeze-out events were observed in the RasGas LNG plant prior to the planned shutdown in August 2014.

4 | CONCLUSIONS

In this work we have presented solid–fluid equilibrium (SFE) and solid formation data for the carbon dioxide + methane binary system at conditions relevant to liquefied natural gas (LNG) production and at carbon dioxide concentrations approaching the industrial specification for LNG. The SFE data are the first to unambiguously describe the solubility of CO₂ in high-pressure liquid methane and extend to

significantly lower concentrations than previously reported. The new data were used to tune the SFE predictions of the model implemented in the ThermoFAST software package, reducing by a factor of 7 the root-mean-square deviation in calculated melting temperatures from measured data.

The solid formation data gathered to quantify the kinetics of formation are, to our knowledge, the first of their kind for this system under industrially relevant conditions. Stationary nucleation rates extracted from the combined 100 and 500 ppm carbon dioxide in methane ramped-temperature datasets agree with those determined from fixed-temperature formation data. Nucleation parameters describing the CO₂ + CH₄ binary system were determined from linearization of the stationary nucleation rates. These parameters were also used to show that formation rates observed at 52 and 66 ppm CO₂ in CH₄ were consistent with those predicted from fits to data measured at higher concentrations.

The measured nucleation rates also imply that the probability of forming a solid within the residence time of fluid in the cryogenic heat exchanger is small unless the concentration of CO₂ is far in excess (~40×) of the industrial specification. Quantification of nucleation rates as a function of process conditions greatly improves the risk assessment of solid formation in industrial systems, allowing qualitative, rule-of-thumb estimates of formation risk and blockage remediation to be replaced by a quantitative, data-driven approach.

AUTHOR CONTRIBUTIONS

Catherine C. Sampson: Data curation (lead); formal analysis (lead); writing – original draft (lead). **Peter J. Metaxas:** Funding acquisition (equal); investigation (equal); project administration (lead); writing – review and editing (equal). **Mark T. J. Barwood:** Data curation (supporting); formal analysis (supporting); writing – original draft (supporting); writing – review and editing (equal). **Rebecca**

Sinclair-Adamson: Data curation (supporting); formal analysis (supporting); writing – review and editing (supporting). **Peter E. Falloon:** Formal analysis (equal); software (lead); writing – original draft (supporting); writing – review and editing (equal). **Paul L. Stanwix:** Funding acquisition (supporting); supervision (equal); writing – review and editing (equal). **Michael Johns:** Funding acquisition (supporting); supervision (supporting); writing – review and editing (equal). **Eric F. May:** Funding acquisition (lead); supervision (equal); writing – review and editing (equal).

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the supplementary material of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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